INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	Allentown-TekPark, PA -- station KTTN, America/New_York
Building	OSM 777196805 Allentown-TekPark
Facade gap	not applicable -- one building, no second facade
Weather record	43,471 real hourly records from KTTN
Plume physics	NOT MODELLED. No other tagged data centre lies inside the solver's validated range, so there is no neighbour intake for a plume to arrive at: the quantity does not exist here rather than being unmeasured. This is a statement about the model's domain, not a claim that recirculation here is zero. A building's own exhaust re-entering its own intake is not modelled at any site in this project.

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2024-10-21 (crossing)
Plant limit	21.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 11 of 24 hours with 2 mode changes. The reactive incumbent took 12 hours -- 1 more -- but declared 1 unsafe hour against the agent's 0. 3 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	11 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
Incumbent	12 of 24 hours free cooling, 2 mode change(s), 1 unsafe hour(s) 3 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	FREE	yes	-	9.440	21.0	9.440	0.773
01	FREE	yes	-	9.440	21.0	9.440	0.682
02	FREE	yes	-	9.160	21.0	8.330	0.879

03	FREE	yes	-	8.605	21.0	7.220	0.822
04	FREE	yes	-	8.330	21.0	6.670	0.753
05	FREE	yes	-	7.500	21.0	6.110	0.701
06	FREE	yes	-	6.945	21.0	5.560	0.751
07	FREE	yes	-	7.780	21.0	6.110	0.969
08	FREE	yes	-	12.500	21.0	13.330	1.498
09	FREE	yes	-	15.555	21.0	17.780	1.930
10	FREE	yes	-	17.220	21.0	20.560	1.797
11	mechanical	no	dry-bulb	21.385	21.0	22.780	1.496
12	mechanical	no	dry-bulb	24.280	21.0	25.560	1.141
13	mechanical	no	dry-bulb	25.615	21.0	26.670	0.885
14	mechanical	no	dry-bulb	26.390	21.0	26.670	0.895
15	mechanical	no	dry-bulb	27.230	21.0	26.670	0.750
16	mechanical	no	dry-bulb	27.225	21.0	26.110	0.928
17	mechanical	no	dry-bulb	26.115	21.0	25.000	0.853
18	mechanical	no	dry-bulb	24.445	21.0	22.220	0.982
19	mechanical	no	dry-bulb	23.890	21.0	22.220	1.006
20	mechanical	no	dry-bulb	22.780	21.0	21.110	1.115
21	mechanical	yes	switch budget	20.555	21.0	18.890	1.325
22	mechanical	yes	switch budget	18.890	21.0	15.560	1.123
23	mechanical	yes	switch budget	18.610	21.0	16.110	0.932

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.440 C, which is 11.560 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.773 C, and the margin is measured, not chosen: 0.773 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

01:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.440 C, which is 11.560 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.682 C, and the margin is measured, not chosen: 0.682 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

02:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.160 C, which is 11.840 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.879 C, and the margin is measured, not chosen: 0.879 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

03:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.605 C, which is 12.395 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.821 C, and the margin is measured, not chosen: 0.821 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

04:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.330 C, which is 12.670 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.753 C, and the margin is measured, not chosen: 0.753 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

05:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.500 C, which is 13.500 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.701 C, and the margin is measured, not chosen: 0.701 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

06:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 6.945 C, which is 14.055 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.751 C, and the margin is measured, not chosen: 0.751 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

07:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.780 C, which is 13.220 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.969 C, and the margin is measured, not chosen: 0.969 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

08:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.500 C, which is 8.500 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.498 C, and the margin is measured, not chosen: 1.498 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

09:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.555 C, which is 5.445 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.930 C, and the margin is measured, not chosen: 1.930 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

10:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.220 C, which is 3.780 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.797 C, and the margin is measured, not chosen: 1.797 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.385 C against a 21.0 C limit, so it fails by 0.385 C. It would take a limit 0.385 C higher, or a bound 0.385 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.280 C against a 21.0 C limit, so it fails by 3.280 C. It would take a limit 3.280 C higher, or a bound 3.280 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.615 C against a 21.0 C limit, so it fails by 4.615 C. It would take a limit 4.615 C higher, or a bound 4.615 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 26.390 C against a 21.0 C limit, so it fails by 5.390 C. It would take a limit 5.390 C higher, or a bound 5.390 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 27.230 C against a 21.0 C limit, so it fails by 6.230 C. It would take a limit 6.230 C higher, or a bound 6.230 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 27.225 C against a 21.0 C limit, so it fails by 6.225 C. It would take a limit 6.225 C higher, or a bound 6.225 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 26.115 C against a 21.0 C limit, so it fails by 5.115 C. It would take a limit 5.115 C higher, or a bound 5.115 C tighter, to change this hour.

18:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.445 C against a 21.0 C limit, so it fails by 3.445 C. It would take a limit 3.445 C higher, or a bound 3.445 C tighter, to change this hour.

19:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.890 C against a 21.0 C limit, so it fails by 2.890 C. It would take a limit 2.890 C higher, or a bound 2.890 C tighter, to change this hour.

20:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.780 C against a 21.0 C limit, so it fails by 1.780 C. It would take a limit 1.780 C higher, or a bound 1.780 C tighter, to change this hour.

21:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 20.555 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

22:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 18.890 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

23:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 18.610 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

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